

Project Proposal: Banking Transaction Analysis

1. Project Overview

1.1 Background

The Banking Transaction Analysis project aims to analyze customer transaction behavior and identify patterns related to transaction activities and fraud risks. By applying data analytics and machine learning techniques, the project supports data-driven decision-making for improving fraud detection, customer understanding, and operational efficiency in the banking sector.

It provides comprehensive information on **transaction records**, **customer profiles**, and **card data** collected from a banking institution throughout the 2010s decade. The dataset reflects real-world banking operations, including customer spending behavior, card usage patterns, and financial interactions across different regions and merchant categories.

In modern banking systems, analyzing transaction data is essential due to:

- The massive volume of daily financial transactions generated across digital and physical channels
- Diverse customer segments with varying income levels, spending behaviors, and financial needs
- Rising risks of **fraud**, **financial crime**, and **abnormal transaction** activities
- The need to improve **customer experience**, **operational efficiency**, and **service quality**
- The growing demand for **personalized financial products**, recommendations, and targeting strategies
- Real-time monitoring of **customer behavior** and transaction anomalies for risk management purposes

1.2 Objectives

The objective is to analyze transactional data to:

- Identify abnormal transaction patterns and suspicious financial behavior
- Understand customer financial behavior across demographic and income segments
- Analyze the most frequently used products and services in transactions.
- Detect transaction anomalies and fraud risks using machine learning models.
- Support banking operations with data-driven insights and visual dashboards.

2. Data Sources

2.1. Dataset:

Banking transaction dataset sourced from: [!\[\]\(3dfb8d66e81160ad61421a3452093d1b_img.jpg\) Financial Transactions Dataset: Analytics](#)

2.2. Data Description:

1. Transaction Data (transactions_data.csv)

- Id: Unique identifier for each transaction.
- date: Timestamp indicating when the transaction occurred.
- client_id: Unique identifier for the customer initiating the transaction.
- card_id: Unique identifier for the payment card used in the transaction.
- amount: Monetary value of the transaction.
- use_chip: Indicates whether the transaction was completed using chip authentication.
- merchant_id: Unique identifier for the merchant processing the transaction.
- merchant_city: City where the merchant is located.
- merchant_state: State or region where the merchant operates.
- zip: ZIP/postal code associated with the merchant location.

2. Card Information (cards_dat.csv)

- Id: Unique identifier for each card record.
- client_id: Unique identifier linking the card to a customer.
- card_brand: Brand of the payment card (e.g., Visa, Mastercard).
- card_type: Type/category of the card such as debit or credit card.
- card_number: Masked or encoded card number associated with the account.
- expires: Expiration date of the payment card.
- cvv: Card verification value used for payment authentication.
- has_chip: Indicates whether the card supports chip-enabled transactions.
- num_cards_issued: Total number of cards issued to the customer.
- credit_limit: Maximum credit limit assigned to the card account.

3. User Data (users_data)

- Id: Unique identifier for each customer.
- current_age: Current age of the customer.
- retirement_age: Expected retirement age of the customer.
- birth_year: Birth year of the customer.
- birth_month: Birth month of the customer.
- gender: Gender of the customer.
- address: Residential address associated with the customer.
- latitude: Geographic latitude of the customer location.

- longitude: Geographic longitude of the customer location.

4. Fraud Labels (train_fraud_labels.json)

- Id: Unique identifier corresponding to each transaction record.
- is_Fraud: Target label indicating whether the transaction is fraudulent (1) or legitimate (0).

3. Project Workflow and Methodology

3.1. Data Collection

- Import transaction dataset into the analysis environment.
- Store and organize the dataset for preprocessing and analysis

3.2. Data Preprocessing

- Handled missing values and duplicate records.
- Generated descriptive statistics to validate data quality and identify inconsistencies.
- Merged multiple datasets into a unified dataframe to support exploratory data analysis (EDA).
- Engineered additional features for analysis, including:
 - flow_type: Categorized transactions as inflow or outflow and converted negative transaction amounts to positive values to better analyze customer cash flow and spending behavior.
 - hour, weekday, is_weekend, month, year: Extracted temporal features from transaction timestamps to capture behavioral and seasonal transaction patterns.
 - days_to_expire: Calculated the remaining days until card expiration to identify potential lifecycle-related transaction behavior.

3.3. Exploratory Data Analysis (EDA)

- Analyze the distribution of categorical features.
- Analyze the distribution of numerical features.
- Analyze customer transaction patterns and fraud distribution.
- Perform RFM analysis and customer segmentation.
- Analyze product usage and transaction trends.
- Perform correlation analysis between important variables.
- Apply Chi-Square test and VIF analysis for feature selection.

3.4. Data Visualization using matplotlib and Power BI

- **Build data visualizations using Matplotlib**
- Visualize:
 - Transaction distribution analysis.

- Fraud rate analysis.
 - RFM analysis and customer segmentation.
 - Product and transaction category analysis.
 - Correlation analysis between key variables.
- **Build interactive dashboards using Power BI**
- Visualize:
 - Executive overview of transaction activities, transaction value, and fraud metrics.
 - Customer segmentation and RFM analysis.
 - Transaction behavior analysis across products and transaction categories.
 - Fraud risk analysis by transaction trends over time, transaction amount, and product usage.
 - Geographic analysis of customer distribution, transaction volume, transaction value, and fraud cases.

3.5. AI Model Development

- Separate numerical and categorical variables.
- Scale numerical features and encode categorical features for machine learning models.
- Transform and vectorize features for model training.
- Split the dataset into training and testing sets.

Train classification models, including Logistic Regression, Random Forest, and GBTCClassifier, for fraud detection.

3.6. Model Evaluation

The fraud detection model was evaluated using the following metrics:

- AUC (Area Under the Curve): Measures the model's ability to distinguish fraudulent and non-fraudulent transactions.
- PR AUC: Evaluates model performance on imbalanced datasets where fraud cases are relatively rare.
- Precision: Measures how many predicted fraud transactions are actually fraud.
- Recall: Measures the model's ability to detect actual fraud cases.
- F1-score: Balances precision and recall for overall fraud detection performance.

4. Insights

- Transaction amount, transaction timing, customer income level, and card usage behavior were identified as the most important factors associated with fraudulent transactions.
- High-value transactions, weekend transactions, dormant accounts, and low-income customer groups showed significantly higher fraud risk compared to other customer segments.

- The analysis revealed a substantial gap between chip card ownership and chip transaction usage, indicating potential security vulnerabilities and opportunities for stronger fraud prevention controls.
- Statistical testing and feature engineering helped identify the most relevant variables for fraud detection while reducing multicollinearity and improving model performance.
- Random Forest and GBTCClassifier achieved higher AUC and PR AUC scores than Logistic Regression, demonstrating that fraud detection benefits from models capable of capturing complex non-linear patterns and feature interactions.
- Random Forest achieved the best overall balance between fraud detection performance and operational efficiency, delivering the highest PR AUC (0.1111) while maintaining an extremely low false positive rate.

5. Recommendations

- Implement risk-based monitoring and multi-factor authentication for high-value transactions and unusual customer activities.
- Strengthen fraud monitoring during weekends and high-risk transaction periods, where fraud rates are significantly higher than normal days.
- Encourage or enforce chip-based transaction usage for chip-enabled cards to reduce security vulnerabilities and improve transaction security.
- Develop customer-specific fraud scoring rules by incorporating transaction amount, customer income level, account activity status, and transaction timing.
- Apply enhanced monitoring and verification procedures for dormant accounts and other high-risk customer segments before allowing large transactions.
- Deploy Random Forest as the primary fraud detection model due to its strong overall performance, highest PR AUC score, and extremely low false positive rate, making it suitable for real-world fraud monitoring and operational efficiency.

6. Tools and Technologies

- Notebook Python (Pandas, PySpark) in Databricks
- Machine Learning models: Logistic Regression, Random Forest, and GBTCClassifier
- Power BI and Python libraries (Matplotlib, Seaborn) for data visualization and dashboard development
- CSV dataset storage and data management using BigQuery

7. Conclusion

This project demonstrates how data analytics and machine learning can support banking transaction monitoring and fraud detection. By analyzing transaction behavior and building predictive models, the project provides valuable insights that can help financial institutions

improve fraud prevention strategies, reduce operational risks, and enhance decision-making processes.